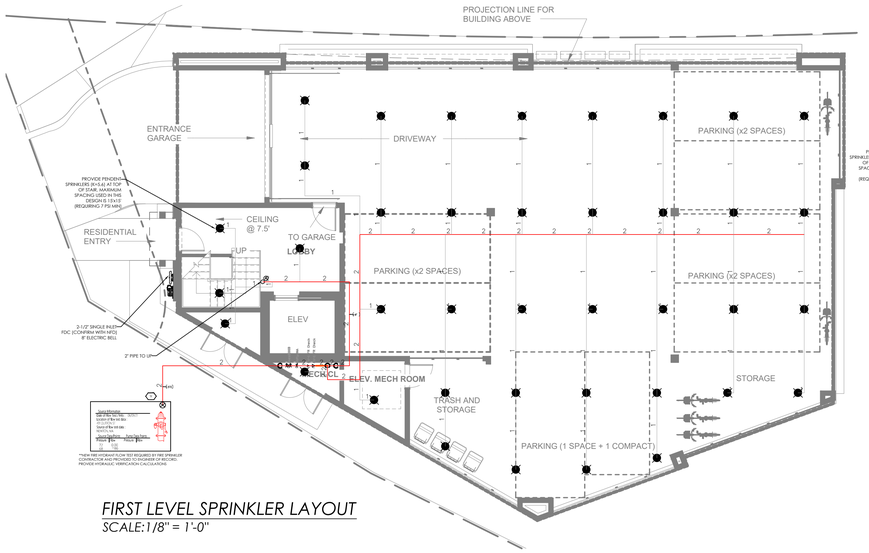
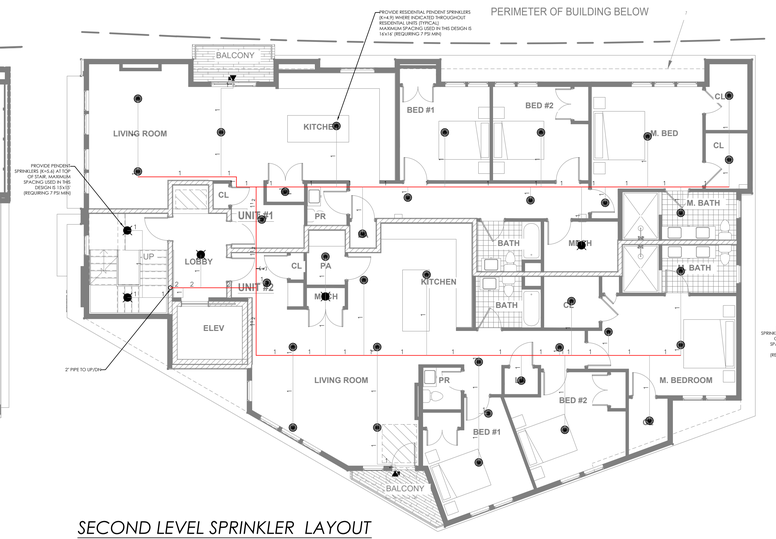
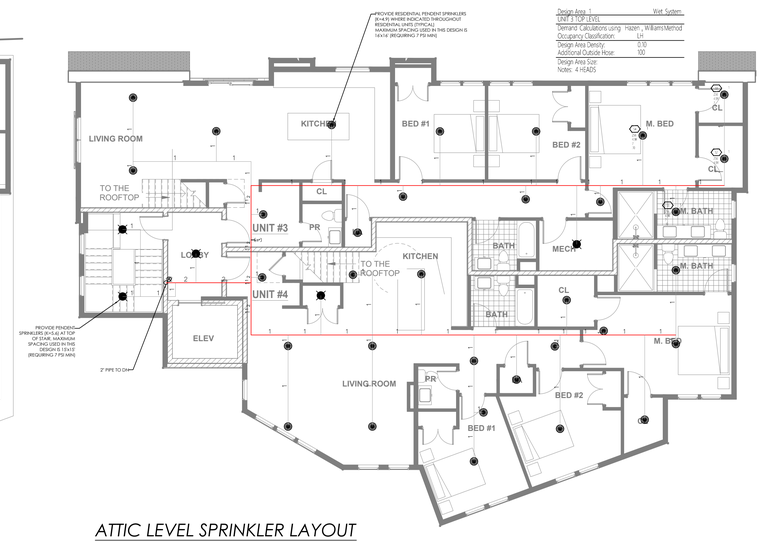


FIRST LEVEL SPRINKLER LAYOUT
SCALE:1/8" = 1'-0"



SECOND LEVEL SPRINKLER LAYOUT
SCALE:1/8" = 1'-0"



ATTIC LEVEL SPRINKLER LAYOUT
SCALE:1/8" = 1'-0"

SPRINKLER PIPE LEGEND

- NEW STEEL PIPING = (SCH. 10 HD STEEL W/ROLL GRVD FITTINGS)
- NEW CPVC PIPING = (CPVC PIPING W/GPVC FITTINGS)
- EXISTING PIPING =
- NEW TAP =

Job:	1062 BEACON ST ST
Address:	1062 BEACON ST ST
City:	NEWTON State: MA Zip/Postal Code: 02459
Contract #:	Date: 03/14/26
Contractor:	CAP DESIGN GROUP Designer: JK
Address:	335 WASHINGTON ST SUITE 1114
City:	WOBURN State: MA Zip/Postal Code: 01801
Phone:	(617) 299 9155 Fax: Email:
Approving Authority:	CITY OF NEWTON

Design Defaults:		Standards:	NFPA13R
Default Sprig Size:	1	Default Drop Size:	1
Default Sprig Material:	CPVC	Default Drop Material:	CPVC
Default Sprig Elevation:	0	Default Drop Elevation:	0

Calculation results for Design Area 1 – UNIT 3 TOP LEVEL

This system as shown on CAP DESIGN GROUP company print no _____ dated 03/14/26 for 1062 BEACON ST ST at 1062 BEACON ST ST contract no _____ is designed to discharge at a rate of 0.1 gpm/ft² (L/min/m²) of floor area over a maximum area of 530 ft² when supplied with water at a rate of 66.5 gpm at 51.6 psi at the base of the riser. Hose stream allowance of _____ is included in the above.

Occupancy classification: LH
Commodity classification: _____
Maximum storage height: _____
Storage arrangement: _____

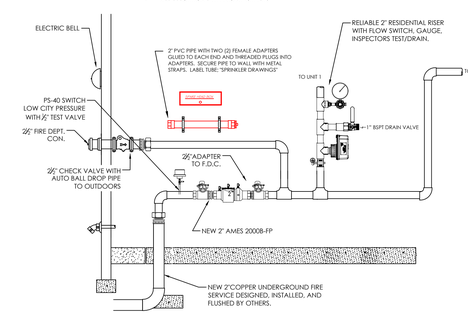
Number of heads flowing: 4
System Type: Wet
Maximum velocity: 22.42 ft/s

Flow from in , Rack sprinklers:	0 gpm
Flow from Overhead sprinklers:	66.5 gpm
Flow from Inside Hoses:	0 gpm
Flow from Outside Hoses:	0 gpm
Other fixed flows:	0 gpm
Total flow in system piping:	66.5 gpm
Additional flow at/beyond source:	100 gpm
Total of all flows:	166.5 gpm

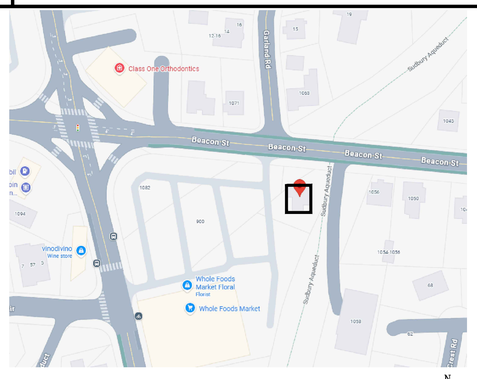
Pressure Required at Source:	51.6 psi
Pressure Available at Source:	71.9 psi
Surplus Pressure at Source:	20.3 psi

4 FIRE SERVICE / BACKFLOW DETAIL

GENERAL NOTES:
- HYDRAULIC PLACARD TO BE INSTALLED ON EACH RISER PER NFPA 13
- ALL ELECTRICAL WIRING BY OTHERS



1 PROJECT INFORMATION



2 GENERAL DETAILS

TABLE 8.6.5.1.2
POSITIONING OF SPRINKLER TO AVOID OBSTRUCTION TO DISCHARGE

DISTANCE FROM SPRINKLERS TO SIDE OF OBSTRUCTION (A)	MAXIMUM ALLOWABLE DISTANCE OF DEFLECTOR ABOVE BOTTOM OF OBSTRUCTION (INCHES) (B)
LESS THAN 1 FT	0
1'-0" TO LESS THAN 1'-4"	0'-3 1/2"
1'-4" TO LESS THAN 2'-0"	0'-5 1/2"
2'-0" TO LESS THAN 2'-4"	0'-7 1/2"
2'-4" TO LESS THAN 3'-0"	0'-11 1/2"
3'-0" TO LESS THAN 3'-4"	0'-8 1/2"
3'-4" TO LESS THAN 4'-0"	1'-0"
4'-0" TO LESS THAN 4'-4"	1'-2"
4'-4" TO LESS THAN 5'-0"	1'-4 1/2"
5'-0" TO LESS THAN 5'-4"	1'-4"
5'-4" TO LESS THAN 6'-0"	1'-8"
6'-0" TO LESS THAN 6'-4"	2'-0"
6'-4" TO LESS THAN 7'-0"	2'-4"
7'-0" TO LESS THAN 7'-4"	2'-11"
7'-4" AND GREATER	1'-7"

NFPA 13 TABLE 8.6.5.1.2 = FIGURE 8.6.5.1.2. (B) STANDARD PENDENT AND UPRIGHT SPRAY SPRINKLERS

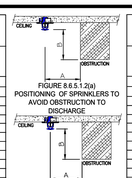


FIGURE 8.6.5.1.2(a) POSITIONING OF SPRINKLERS TO AVOID OBSTRUCTION TO DISCHARGE

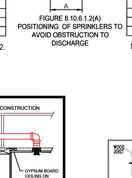


FIGURE 8.10.6.1.2(A) POSITIONING OF SPRINKLERS TO AVOID OBSTRUCTION TO DISCHARGE

TABLE 8.10.6.1.2
POSITIONING OF SPRINKLER TO AVOID OBSTRUCTION TO DISCHARGE

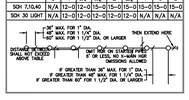
DISTANCE FROM SPRINKLERS TO SIDE OF OBSTRUCTION (A)	MAXIMUM ALLOWABLE DISTANCE OF DEFLECTOR ABOVE BOTTOM OF OBSTRUCTION (INCHES) (B)
LESS THAN 1 FT	0
1'-0" TO LESS THAN 1'-4"	0'-6"
1'-4" TO LESS THAN 2'-0"	0'-11"
2'-0" TO LESS THAN 2'-4"	0'-11"
2'-4" TO LESS THAN 3'-0"	0'-11"
3'-0" TO LESS THAN 3'-4"	0'-2"
3'-4" TO LESS THAN 4'-0"	0'-3"
4'-0" TO LESS THAN 4'-4"	0'-5"
4'-4" TO LESS THAN 5'-0"	0'-7"
5'-0" TO LESS THAN 5'-4"	0'-7"
5'-4" TO LESS THAN 6'-0"	0'-7"
6'-0" TO LESS THAN 6'-4"	0'-9"
6'-4" TO LESS THAN 7'-0"	0'-11"
7'-0" AND GREATER	1'-2"

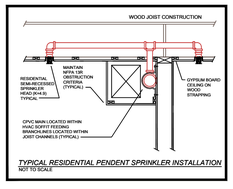
NFPA 13 TABLE 8.10.6.1.2 = FIGURE 8.10.6.1.2 (A) RESIDENTIAL PENDENT AND UPRIGHT SPRAY SPRINKLERS

NOTE: AT THE TIME OF THIS SPRINKLER SYSTEM DESIGN, THE RCP AND LIGHTING SELECTION HAD NOT BEEN COMPLETED. SPRINKLER HEADS HAVE BEEN POSITIONED IN ACCORDANCE WITH CODE REQUIREMENTS AND HAVE NOT BEEN COORDINATED WITH LIGHTING OR OTHER SURFACE-MOUNTED FIXTURES. ANY LIGHTS OR OTHER FIXTURES THAT CAUSE OBSTRUCTIONS TO SPRINKLER HEAD SPRAY PATTERNS WILL REQUIRE ADDITIONAL SPRINKLER HEADS AT AN ADDITIONAL COST TO THE BUILDING OWNER.

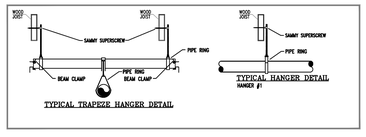
MAX. HANGER SPACING

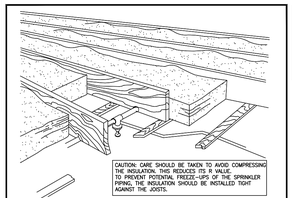
PIPE TYPE	NOMINAL PIPE SIZE (IN)							
	1	1 1/4"	1 1/2"	2	2 1/2"	4	6	8
SCH 7,10,40	N/A	12-0	12-0	15-0	15-0	15-0	15-0	15-0
SCH 30 LIGHT	N/A	12-0	12-0	12-0	12-0	12-0	N/A	N/A





TYPICAL RESIDENTIAL PENDENT SPRINKLER INSTALLATION
NOT TO SCALE

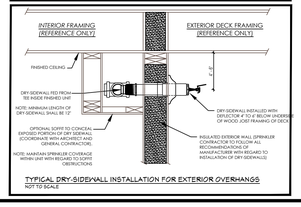




INSULATION PROTECTING SPRINKLER PIPING ON UPPER FLOOR DETAIL
NOT TO SCALE

NOTE: IT IS THE BUILDING OWNER'S RESPONSIBILITY TO PROVIDE SUFFICIENT HEAT WITHIN ALL AREAS CONTAINING SPRINKLER PIPING TO PREVENT FREEZE-UPS. CAP DESIGN GROUP, INC. TAKES NO RESPONSIBILITY FOR FREEZE-UPS ON THIS SPRINKLER SYSTEM. ANY AREAS OF CONCERN SHALL BE BROUGHT TO THE ENGINEER'S ATTENTION IN WRITING PRIOR TO INSTALLATION.

NOTE: ALL COMBUSTIBLE CONCEALED SPACES, INCLUDING INTERSTITIAL SPACES, VOIDS UNDER STAIRS, ETC., WHICH ARE GREATER THAN 6" IN DEPTH, SHALL BE COMPLETELY FILLED WITH NON-COMBUSTIBLE INSULATION BY THE GENERAL CONTRACTOR OR ARE REQUIRED TO BE PROVIDED WITH SPRINKLER PROTECTION IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13 (2019 EDITION). ANY SUCH ARES CREATED DURING CONSTRUCTION SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER IN WRITING BY THE GENERAL CONTRACTOR PRIOR TO INSTALLATION OF GYPSUM BOARD WALLS AND/OR CEILINGS.



TYPICAL DRY-SIDEWALL INSTALLATION FOR EXTERIOR OVERHANGS
NOT TO SCALE

3 GENERAL NOTES

FIRE PROTECTION NOTES:

THE PURPOSE OF THIS FIRE PROTECTION DRAWING AND THE ASSOCIATED FIRE PROTECTION DESIGN NARRATIVE IS TO INDICATE THE PROPOSED RESIDENTIAL SPRINKLER SYSTEM TO BE INSTALLED WITHIN THE MULTI-UNIT BUILDING LOCATED AT **1062 BEACON ST, NEWTON, MA 02459.**

THIS BUILDING IS COMPRISED OF A BASEMENT LEVEL WITH THREE RESIDENTIAL FLOORS ABOVE. THE BUILDING IS LESS THAN 30-FT ABOVE FIRE DEPARTMENT VEHICULAR ACCESS AND THEREFORE DOES NOT REQUIRE A STANDPIPE SYSTEM.

THIS RESIDENTIAL SPRINKLER SYSTEM HAS BEEN DESIGNED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13R (2019 EDITION) FOR A RESIDENTIAL BUILDING UP TO 3 STORIES IN HEIGHT AND NOT EXCEEDING 12,000 GROSS SF.

THE SPRINKLER CONTRACTOR SHALL FOLLOW THE LATEST REQUIREMENTS OF NFPA 13R (2019 EDITION), MASSACHUSETTS STATE BUILDING CODE (780 CMR)10TH EDITION (IBC-2021) AND **NEWTON** FIRE DEPARTMENT REQUIREMENTS.

THIS SYSTEM WILL BE SUPPLIED BY A NEW 2" K TYPE FIRE SERVICE FLUSHED AND TESTED AND MAINTAINED PER NFPA 25. FIRE SPRINKLER CONTRACTOR SHALL WITNESS THE FLUSH AND TEST.

THE SYSTEM HAS BEEN HYDRAULICALLY CALCULATED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13R, INCLUDING THE FOUR HYDRAULICALLY MOST DEMANDING HEADS IN A SINGLE COMPARTMENT BASED ON THE REQUIREMENTS OF THE SPECIFIC SPRINKLER HEAD AND THE SPACING USED IN THIS DESIGN (16'x16' REQUIRING A MINIMUM OF 13 GPM @ 7.0 PSI). REMOTE AREAS, DENSITIES AND HOSE STREAM ALLOWANCES ARE INDICATED ON THIS DRAWING ALONG WITH SYSTEM DEMANDS AT CONNECTION TO STREET. CALCULATION RESULTS HAVE BEEN COMPARED TO RECENT HYDRANT FLOW TEST INFORMATION OBTAINED FROM THE **NEWTON** WATER & SEWER DEPARTMENT. CALCULATIONS HAVE BEEN SUBMITTED WITH THIS DRAWING TO THE **NEWTON** FIRE DEPARTMENT FOR REVIEW.

INSTALLATION NOTES:

ALL WORK SHALL BE PERFORMED BY A MASSACHUSETTS LICENSED SPRINKLER CONTRACTOR. THE SPRINKLER CONTRACTOR SHALL FOLLOW THE LATEST REQUIREMENTS OF NFPA 13R (2019 EDITION), MASSACHUSETTS STATE BUILDING CODE (780 CMR)10TH EDITION (IBC-2021) AND THE **NEWTON** FIRE DEPARTMENTS.

THE ARCHITECTURAL BACKGROUND OF BUILDING MAY DIFFER SLIGHTLY FROM ACTUAL LAYOUT. DRAWINGS ARE NOT INTENDED TO SHOW ALL OFFSETS AND PIPING ELEVATION CHANGES. CONTRACTOR SHALL FIELD VERIFY ALL MEASUREMENTS PRIOR TO FABRICATION.

CONTRACTOR SHALL HYDROSTATICALLY TEST ALL SPRINKLER PIPING AT 200 PSI FOR 2 HOURS AND IS RESPONSIBLE FOR THE COMPLETION OF ALL ABOVE GROUND TEST CERTIFICATES, SUPPLIED TO THE OWNER.

ALL PIPING INSTALLED THROUGHOUT THE RESIDENTIAL AREAS OF THE BUILDING SHALL BE UL LISTED CPVC SPRINKLER PIPING. ALL PIPING WITHIN THE STORAGE AND UTILITY AREAS OF THE BASEMENT LEVEL WILL BE BLACK STEEL PIPING. ALL PIPING SHALL BE INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 13R(2019 EDITION) AND ALL MANUFACTURERS INSTALLATION RECOMMENDATIONS. ALL PIPING SHALL BE PITCHED TO DRAIN WITH LOW-POINT DRAINS AT SECTIONS OF PIPING SUBJECT TO WATER TRAPPING. SPRINKLER CONTRACTOR SHALL PROVIDE PROTECTIVE PLATES WHERE CPVC PIPING IS RUN THROUGH STUDS TO PREVENT PUNCTURING OF THE SPRINKLER PIPING DURING DRYWALL INSTALLATION AS REQUIRED BY NFPA STANDARDS.

ALL SPRINKLER HEADS WITHIN RESIDENTIAL AREAS OF THE BUILDING SHALL BE RESIDENTIAL PENDENT AND SIDEWALL SPRINKLERS. SPRINKLER HEADS WITHIN THE BASEMENT UTILITY AREAS SHALL BE QUICK-RESPONSE COMMERCIAL UPRIGHT SPRINKLERS. ALL HEADS SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S REQUIREMENTS AND THE REQUIREMENTS OF NFPA 13R (2019 EDITION).

BUILDING OWNER IS RESPONSIBLE FOR MAINTAINING THIS SPRINKLER SYSTEM IN ACCORDANCE WITH THE REQUIREMENTS OF NFPA 25, INCLUDING THE PROVISION OF HEAT IN ALL AREAS CONTAINING SPRINKLER PIPING AND HEADS TO PREVENT PIPE FROM FREEZING. CAP DESIGN GROUP AND THE ENGINEER OF RECORD TAKE NO RESPONSIBILITY FOR ANY DAMAGES CAUSED BY FREEZE UPS.

PERMIT NO.	
CONTRACT NO.	
APPROVAL	
DRAWN BY	JK
SCALE	1/8" = 1'-0"
DATE	03/24/26
SHEET NUMBER	FP 1 of 1

FIRE PROTECTION TIER 1

1062 BEACON ST
NEWTON, MA 02459

ENGINEER CAP DESIGN GROUP

CAP DESIGN GROUP
www.capcofire.com
335 WASHINGTON ST SUITE 1114
WOBURN, MA 01801
(617) 299 9155

CAP DESIGN GROUP

ENGINEER'S STAMP:

SPRINKLER SYMBOLS

SYM	QNT	POSITION	FINISH	TEMP	K	NPT	NPS	MODEL #
	49	PEND	WHITE	155	5.60	1/2"	Victaulic	V27
	68	RES PEND	WHITE	175	4.90	1/2"	Victaulic	F-RES
	2	SIDE	WHITE	155	5.60	1"	Victaulic	F-RES

REVISIONS		
DATE	DESCRIPTION	BY